

File Integrity Monitoring:

File Integrity Monitoring is to monitor any changes to the file like creating, modifying and deleting file in the root directory.

What we will do?

We will create a file in **Ubuntu** and **Windows** and then do some modifications in it. After then we'll see if we can see events generated by modifying file in the agents.

FIM on Ubuntu:

- Click on the Ubuntu agent > File Integrity Monitoring > Events

The screenshot shows the Wazuh dashboard interface. At the top, there's a navigation bar with 'Endpoints' and 'ubuntu-agent' selected. Below this, a menu bar includes 'Threat Hunting', 'File Integrity Monitoring' (highlighted with a red box), 'Configuration Assessment', 'MITRE ATT&CK', and 'More...'. A table below lists agent details: ID 004, Status active, IP address 192.168.23.131, Version Wazuh v4.12.0, Group default, Operating system Ubuntu 24.04.2 LTS, and Cluster node node01. Below the table, there's another navigation bar with 'Dashboard', 'Inventory', and 'Events' (highlighted with a red box). A search bar and filters are present, showing 'manager.name: wazuh-server', 'rule.groups: syscheck', and 'agent.id: 004'. A message at the bottom states 'No results match your search criteria'.

- We can see that we do not have any event/alerts yet

Let's go to the Ubuntu machine and create a directory and file

- Use this command:

sudo mkdir -p /opt/fim-demo (It will create a directory)

echo "start" | sudo tee /opt/fim-demo/watchme.txt

(It will create file named watchme.txt and save the word start in it)

```
zayan@zayan:~$ sudo mkdir -p /opt/fim-demo
echo "start" | sudo tee /opt/fim-demo/watchme.txt
```

```

root@zayan:/home# cd /opt
root@zayan:/opt# ls
fim-demo
root@zayan:/opt#

```

Basically, this command will first create a directory **fim-demo** and then save the word **start** in the file **watchme.txt** by creating it.

- Now open the configuration file by using this command

sudo nano /var/ossec/etc/ossec.conf

```

GNU nano 7.2 /var/ossec/etc/ossec.conf
!--
Wazuh - Agent - Default configuration for ubuntu 24.04
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>
  <client>
    <server>
      <address>192.168.10.10</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
    <config-profile>ubuntu, ubuntu24, ubuntu24.04</config-profile>
    <notify_time>10</notify_time>
    <time-reconnect>60</time-reconnect>
    <auto_restart>yes</auto_restart>
    <crypto_method>aes</crypto_method>
    <enrollment>
      <enabled>yes</enabled>
      <agent_name>ubuntu-agent</agent_name>
      <groups>default</groups>
      <authorization_pass_path>etc/authd.pass</authorization_pass_path>
    </enrollment>
  </client>

  <client_buffer>
    <!-- Agent buffer options -->
    <disabled>no</disabled>
    <queue_size>5000</queue_size>
    <events_per_second>500</events_per_second>
  </client_buffer>

  <!-- Policy monitoring -->
  <rootcheck>
    <disabled>no</disabled>
    <check_files>yes</check_files>
    <check_trojans>yes</check_trojans>
  </rootcheck>
</ossec_config>

```

- Locate the **<syscheck>** block in this file and add this line in it

<directories realtime="yes" report_changes="yes">/opt/fim-demo</directories>

You can add multiple lines for multiple directories which you want to monitor

```

<!-- File integrity monitoring -->
<syscheck>
  <disabled>no</disabled>
  <directories realtime="yes" report_changes="yes">/opt/fim-demo</directories>

```

- Save the file and exit (CTRL + O > Enter > CTRL + X)
- Restart the Wazuh agent using this command

sudo systemctl restart wazuh-agent

- Use this command to test the changes in the file

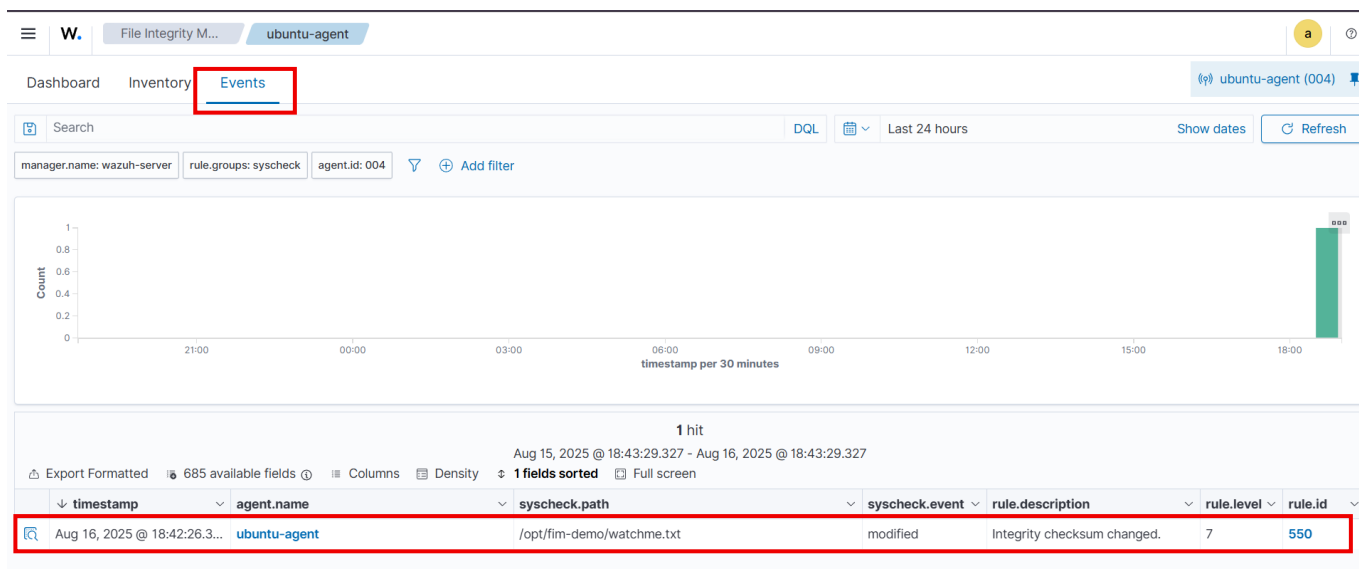
echo "changed \$(date)" | sudo tee -a /opt/fim-demo/watchme.txt

This command will add word changed with current date and save it to the watchme.txt file without overwriting the previous line.

```
root@zayan:/opt# cat /opt/fim-demo/watchme.txt
start
changed Sat Aug 16 01:42:26 PM UTC 2025
root@zayan:/opt#
```

Now check events and dashboard and create or modify the logs.

- Go to the Ubuntu-agent in Wazuh > File Integrity Monitoring > Events



Now here we can see the events (what happened to our file etc.)

We can see everything like which file is modified (if we've added more paths or more directories), when it is modified and other details.

Document Details

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[Table](#) [JSON](#)

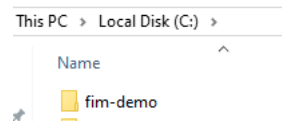
t _index	wazuh-alerts-4.x-2025.08.16
t agent.id	004
t agent.ip	192.168.23.131
t agent.name	ubuntu-agent
t decoder.name	syscheck_integrity_changed
t full_log	File '/opt/fim-demo/watchme.txt' modified Mode: realtime Changed attributes: size,mtime,md5,sha1,sha256 Size changed from '6' to '46' Old modification time was: '1755350377', now it is '1755351746' Old md5sum was: '889e0d7343405c079195e7b8903c8c9e' New md5sum is: '2054442165c28bd53a8a33aa6a9d502b'
t id	1755351746.2854148
t input.type	log
t location	syscheck
t manager.name	wazuh-server
t rule.description	Integrity checksum changed.
# rule.firedtimes	1
t rule.gdpr	II_5.1.f
t syscheck.gname_after	root
t syscheck.inode_after	393495
t syscheck.md5_after	825d4d2165c28bd53a8a33aa6a9d502b
t syscheck.md5_before	889e0d7343405c079195e7b8903c8c9e
t syscheck.mode	realtime
syscheck.mtime_after	Aug 16, 2025 @ 18:42:26.000
syscheck.mtime_before	Aug 16, 2025 @ 18:19:37.000
t syscheck.path	/opt/fim-demo/watchme.txt
t syscheck.perm_after	rw-r--r--
t syscheck.sha1_after	d49727ddfef0467059aaa931341873f7d2eb427a
t syscheck.sha1_before	f6fd1656d79b06918bb5cf85c8bae58c3884deaf
t syscheck.sha256_after	8e4335e86cccc074d9363863c844ce541aceebb1857ee915cb61efe75958f39
t syscheck.sha256_before	46210dddc66714c3d8d226711510cf8421774214016c508c72a833a05370f6b5
# syscheck.size_after	46
# syscheck.size_before	6
t syscheck.uid_after	0
t syscheck.uname_after	root
timestamp	Aug 16, 2025 @ 18:42:26.334

FIM on Windows:

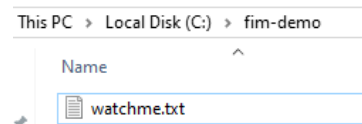
Similarly, Go to your Windows machine

1. Create a test folder and file

- Open C:\ and create a new folder



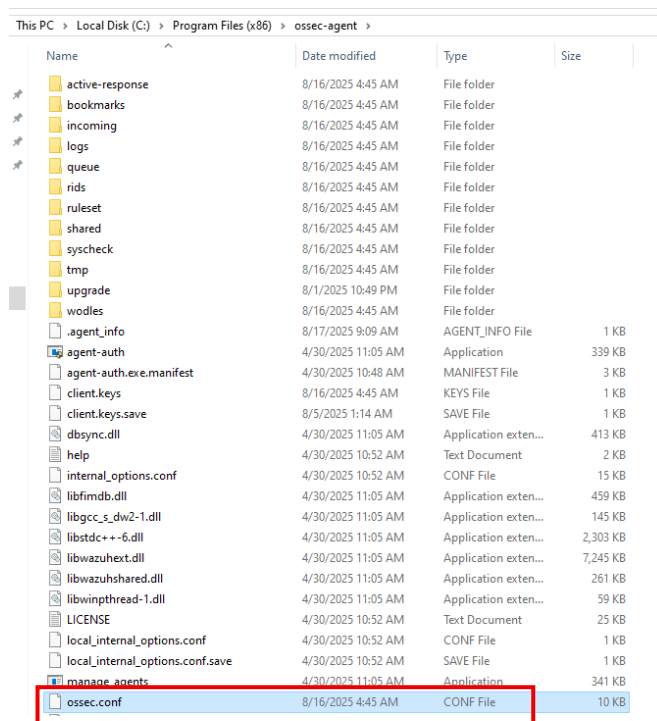
- Inside it, create a file watchme.txt and write some text (e.g., start).



2. Edit the Wazuh agent configuration

- Open this file in Notepad as Administrator:

C:\Program Files (x86)\ossec-agent\ossec.conf



- Look for the <syscheck> block.
If it exists → add this line inside.

<directories realtime="yes" report_changes="yes">C:\fim-demo</directories>

```
<!-- File integrity monitoring -->
<syscheck>

<disabled>no</disabled>
<directories realtime="yes" report_changes="yes">C:\fim-demo</directories>

<!-- Frequency that syscheck is executed default every 12 hours -->
<frequency>43200</frequency>
```

3. Restart the Windows Wazuh agent

- Open PowerShell (Run as Administrator) and restart the service:

Restart-Service -Name WazuhSvc

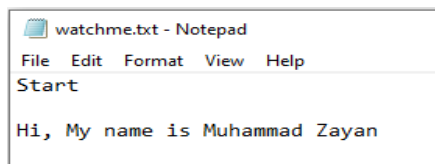
```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/powershell

PS C:\Windows\system32> Restart-Service -Name WazuhSvc
PS C:\Windows\system32>
```

4. Test file monitoring

- Go to C:\fim-demo\watchme.txt
- Add or modify some text (e.g., Hi, My name is Muhammad Zayan)

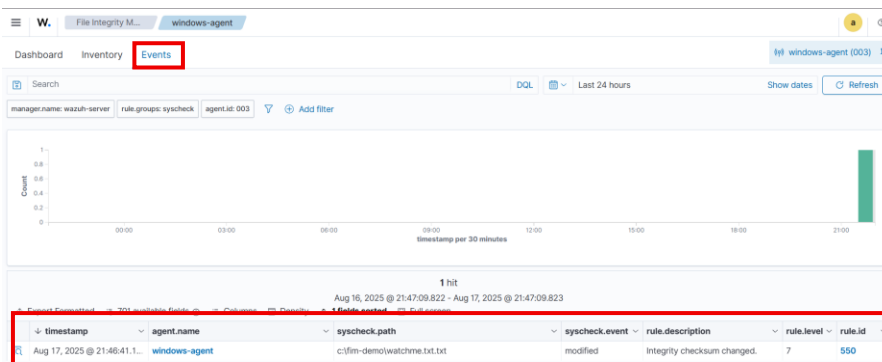


- Save the file

5. Verify in Wazuh Dashboard

- Navigate to **File Integrity Monitoring** → **Events**

You can see the events which happened to the file.



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Table JSON

t _index	wazuh-alerts-4.x-2025.08.17
t agent.id	003
t agent.ip	192.168.23.128
t agent.name	windows-agent
t decoder.name	syscheck_integrity_changed
t full_log	File 'c:\fim-demo\watchme.txt.txt' modified Mode: realtime Changed attributes: size,mtime,md5,sha1,sha256 Size changed from '5' to '38' Old modification time was: '1755448600', now it is '1755449200' Old md5sum was: 'a6122a65eaa676f700ae68d393054a37' New md5sum is: '8376e4b0e5270821e6566305dd9f9df'
t id	1755449201.345594
t input.type	log
t location	syscheck
t manager.name	wazuh-server
t rule.description	Integrity checksum changed.
# rule.firedtimes	1
t rule.gdpr	II_5.1.f

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t rule.mitre.tactic	Impact
t rule.mitre.technique	Stored Data Manipulation
t rule.nist_800_53	SI.7
t rule.pci_dss	11.5
t rule.tsc	PI1.4, PI1.5, CC6.1, CC6.8, CC7.2, CC7.3
① syscheck.attrs_after	ARCHIVE
t syscheck.changed_attributes	size, mtime, md5, sha1, sha256
t syscheck.diff	--- > > Hi, My name is Muhammad Zayan
t syscheck.event	modified
t syscheck.md5_after	8376e4b0e5270821e6566305dd9f9df
t syscheck.md5_before	a6122a65eaa676f700ae68d393054a37
t syscheck.mode	realtime
📅 syscheck.mtime_after	Aug 17, 2025 @ 21:46:40.000
📅 syscheck.mtime_before	Aug 17, 2025 @ 21:36:40.000
t syscheck.path	c:\fim-demo\watchme.txt.txt
t syscheck.sha1_after	40df9b3cf9ad594e65fef0619931b5befd162d02
t syscheck.sha1_before	952f375412e89ff213a8aca383d18e5691354347

